MIS: Diagnosing Structural Bottlenecks in Copy-Paste Augmentation for Agricultural Object Detection

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***Abstract*—Copy-paste augmentation achieves strong results for instance segmentation yet yields inconsistent or negative gains for bounding-box detection in agricultural scenes. We propose the Minimum Inference Score (MIS), a difficulty metric derived from a trained baseline detector that correlates strongly with detection failure modes—miss rate and false-positive rate—across objects and images. MIS partition analysis reveals where and how a model fails: hard instances concentrating in structurally constrained regimes—driven by object size, domain composition, or both—signal that copy-paste is unlikely to provide reliable gains, as confirmed across all tested selection strategies. We validate this diagnostic across three agricultural benchmarks, where MIS partition analysis also predicts relative selection strategy risk from partition composition, providing bidirectional validation across constrained and unconstrained datasets.**

***Index Terms*—object detection, copy-paste augmentation, difficulty scoring, agricultural object detection, data augmentation diagnostic**

I. Introduction

Copy-paste augmentation composites cropped instances into training images to increase diversity [1], [2], yet yields inconsistent or negative gains for bounding-box detection in agricultural imagery subject to sub-resolution instances, multi-domain acquisition, and high object density—forcing costly multi-seed search without a principled stopping criterion. Critically, across all tested selection strategies and datasets with structural constraints, copy-paste consistently fails to provide reliable gains under YOLOv8s, suggesting the bottleneck is structural rather than a sampling problem.

We introduce the **Minimum Inference Score (MIS)**, a metric computed from a trained baseline detector that captures model failure patterns on a given dataset, enabling two practical outputs without any augmented run: determining whether copy-paste can provide useful training signal, and predicting relative selection strategy risk from partition composition alone. On MinneApple [3], GWHD 2021 [4], and WGISD [5], MIS partition analysis identifies structural constraints that predict copy-paste failure and positive gains in their absence, providing bidirectional diagnostic validation.

II. Methodology

A. Scoring Formulation

Let $G(I)$ denote ground-truth boxes in image I. A trained detector produces predictions p with confidence c_p at $\tau_{\text{conf}}{=}0.001$ to retain near-miss candidates. For each $g \in G(I)$, let $\mathcal{P}(g){=}\{p : \text{IoU}(p,g){\geq}0.5\}$. The *Object Difficulty Score* is

$$S_{\text{obj}}(g) = \begin{cases} \min_{p\in\mathcal{P}(g)}[\alpha(1{-}c_p) + \beta(1{-}u_p)], & \mathcal{P}(g) \neq \varnothing \\ \alpha + \beta, & \mathcal{P}(g) = \varnothing \end{cases} \tag{1}$$

with $\alpha{=}\beta{=}0.5$ and $u_p{=}\text{IoU}(p,g)$. Unmatched objects receive the maximum score ($\alpha{+}\beta{=}1$). The *Image Difficulty Score* is

$$S_{\text{img}}(I) = \frac{1}{|G(I)|}\sum_{g} S_{\text{obj}}(g) + w_2 \cdot \text{FPrate}(I), \tag{2}$$

where $\text{FPrate}(I)$ counts unmatched predictions per image. This yields $S_{\text{obj}} \in [0,1]$ for every ground-truth object and S_{img} for every training image, where higher values indicate greater detection difficulty. By construction, S_{img} encodes both axes of detection failure (miss rate and FP rate); w_2 is set by grid search to balance their contributions. MIS scores are model-conditioned and reflect structural properties through detector failure behavior, not directly.

B. Augmentation Protocol

We train a YOLOv8s baseline and partition all training objects by S_{obj} and all training images by S_{img} into equal difficulty tertiles (Low, Medium, High). Four copy-paste conditions are evaluated: **Score High** pastes objects from the highest-difficulty object tertile into the highest-difficulty destination images; **Score Medium** and **Score Low** follow the same logic for their respective tertiles. **Random** selects both destination images and source objects without score-based filtering. This design isolates whether targeting specific difficulty regimes affects outcome. Training data is expanded by 30% via unblended copy-paste [1]: 2–8 objects per image, scale $[0.9, 1.1]$, rotation $\pm 15°$, $3\times$ maximum reuse; all other settings follow default YOLOv8 protocol.

III. Results

A. MIS Reflects Detection Failure Modes

Pearson correlations between S_{img} and miss rate on the validation set are $r{=}0.79$, 0.81, 0.91 on MinneApple, GWHD, and WGISD; correlations with FP rate are $r{=}0.79$, 0.81, 0.84. Strong correlations across all three datasets confirm that MIS captures the primary axes of detection failure as intended by its construction.

TABLE I

MINNEAPPLE: S_{obj} CAPTURES SIZE-LEVEL DIFFICULTY.

Category	AP	Mean S_{obj}	Top tertile
Very Small ($\leq$400 px^2)	0.071	0.152	64.4%
Small (400–1024 px^2)	0.270	0.076	33.9%
Medium ($>$1024 px^2)	0.493	0.053	1.7%

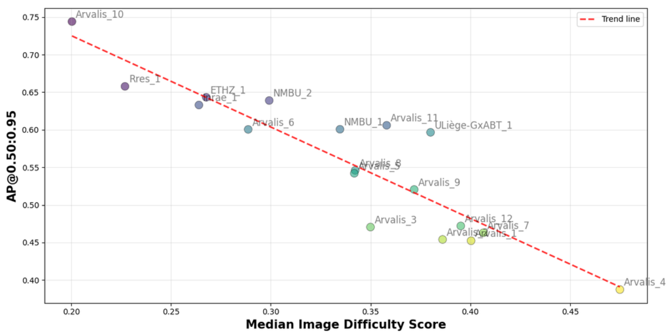

Fig. 1. GWHD 2021: per-domain AP vs. median image difficulty (MIS).

B. MIS Identifies Structural Failure Sources

Size (MinneApple). S_{obj} inversely tracks object area across all training instances (Spearman r_s=-0.82; Table I): 64.4% of the top-difficulty tertile falls in *Very Small* instances ($\leq$400 px^2), where AP is near-zero (0.071), indicating hard objects concentrate below the model's size recognition threshold.

Domain (GWHD 2021). Per-domain median MIS strongly anticorrelates with domain AP (Pearson r=-0.90; Fig. 1): MIS correctly ranks domain difficulty from detector behavior alone, regardless of its underlying cause—varied cameras, growth stages, and geographic acquisition conditions.

C. Application: Pre-Augmentation Diagnostic for Copy-Paste

MinneApple (resolution ceiling). MIS partition analysis identifies hard objects concentrated below the recognition threshold. All four copy-paste conditions yield negative ΔAP (Table II), confirming that additional sub-resolution instances provide no usable learning signal regardless of selection strategy.

GWHD 2021 (domain difficulty). MIS partitioning reveals near-complete domain separation: low-difficulty images are 66% in ETHZ_1/Rres_1 domains (AP$\approx$0.65), while high-difficulty images are 78% Arvalis variants (AP$\approx$0.46). This structure directly predicts per-condition behavior. Score Low sources from high-AP domains, reinforcing already well-learned patterns, resulting in consistent low-variance drop (-0.034, σ=0.005). Score High sources from hard Arvalis domains where difficulty is acquisition-based; added instances cannot resolve domain gaps, yielding high instability (-0.006, σ=0.040). Random and Medium sample without domain awareness, producing moderate drops (-0.015, -0.017) at intermediate variance (σ=0.023, 0.021). Both outcome direction and variance scale with sourced-domain difficulty—patterns

TABLE II

COPY-PASTE OUTCOMES (3-SEED MEAN $\pm\,\sigma$). BOTTLENECK: STRUCTURAL FAILURE SOURCE IDENTIFIED BY MIS.

Dataset	Bottleneck	Base AP	Best Δ	Worst Δ
MinneApple	Size	0.353 ± 0.005	-0.003 (Hi)	-0.008 (Rnd)
GWHD 2021	Domain	0.229 ± 0.013	-0.006^* (Hi)	-0.034 (Lo)
WGISD	—	0.480 ± 0.007	$+0.009$ (Med)	$+0.001$ (Hi)

*Score High on GWHD: σ=0.040, high seed variance.

MIS predicts from partition composition alone, confirming a structural bottleneck no selection strategy can overcome.

WGISD (positive control). MIS raises no flag; all conditions confirm viability with positive ΔAP (Table II), with Score Medium achieving the best result.

Across both flagged datasets, all selection strategies yield negative ΔAP—selection-invariant failure that implicates structural constraints rather than sampling policy as the root cause, and directly explains why copy-paste struggles in detection when such constraints are present.

IV. CONCLUSION

MIS links baseline detector failure behavior to difficulty concentration patterns, enabling two practical outputs: a pre-augmentation diagnostic for copy-paste viability, and a predictor of relative selection strategy risk from partition composition alone. The consistent failure across all selection strategies on structurally constrained datasets indicates that copy-paste's unreliability in detection stems from dataset structure, not selection policy—a distinction MIS partition analysis makes visible before any augmented run. In our experiments, 24 augmented training runs across MIS-flagged datasets (MinneApple and GWHD, 4 conditions $\times$ 3 seeds each) yielded zero reliable gains; MIS identifies these datasets without any augmented run, directly saving that compute budget. Practitioners should apply MIS after baseline training: inspecting partition composition for concentrated high-difficulty instances reveals structural constraints under which copy-paste augmentation is unlikely to yield reliable gains. Future work includes using MIS to guide instance-level selection when structural constraints are absent, and combining MIS with generative methods to produce information-rich training instances for hard regimes.

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